

## Pokhara University

**Level:** Bachelor

**Semester:** VI

**Year:** [2025]

**Programme:** BE IT

**Full Marks:** 100

**Pass Marks:** 45

**Course:** Data Science and Analytics

**Time:** 3 hrs

**Candidates are required to give their answers in their own words as far as practicable.**

**The figures in the margin indicate full marks.**

**Attempt all the questions.**

1. a) Define the Data Analysis Pipeline. Explain the steps involved in data preprocessing with examples. [7]  
b) Differentiate between structured and unstructured data with suitable examples. [8]
2. a) Define and explain numeric and categorical variables. Provide two examples of each. [8]  
b) What is empirical distribution? Illustrate its use for numeric and categorical data. [7]
3. a) Explain the concept of correlation analysis. How do Pearson correlation and cross-tabulation work? [8]  
b) What is statistical significance? Explain p-value and chi-square test with examples. [7]
4. a) Explain multivariate linear regression with an example. What is the role of R-squared and Adjusted R-squared? [8]  
b) Define Principal Component Analysis (PCA). Explain how PCA helps in dimensionality reduction. [7]
5. a) Describe non-parametric regression using the Nadaraya-Watson method. [7]  
b) What is the importance of bandwidth in kernel regression? How is it selected? [8]
6. a) Explain logistic regression and its application in classification problems. [8]  
b) Define ARMA and ARIMA models. What is the role of AIC and BIC in model selection? [7]
7. **Write short notes on any two of the following:** [2 × 5 = 10]
  - a) Granger Causality
  - b) Variable Importance using permutation test
  - c) Zipf's Law